GATE

CS & IT

Discrete Mathematics

DPP: 3

Set Theory & Algebra

Q1 Let A= {a, b, c, d, e}.
Cardinality of the equivalence relation on set A induced by partition {{a, c}, {d}, {b, e}} of set A is________

Q2 Let A is a finite set such that |A|= 5, then cardinality of largest partial order relation possible on set A is______________

Q3 Let A = {a, b, c, d}
The number of total order relations possible on set A are__________

Q4 Let A = {1, 2, 3, 4, 5, 6}.
The number of partitions of set A such that equivalence relations induced by those partitions will contain exactly 12 ordered pairs are____________

Q5 Let R and S are two equivalence relation on set A. Which of the following is true?
(A) R ∪ S is also an equivalence relation on set A.
(B) R ∩ S is also an equivalence relation on set A.
(C) Both R ∪ S as well as R ∩ S are equivalence relation on set A.
(D) Neither R ∪ S nor R ∩ S are equivalence relation on set A.

Q6 Consider the following POSET
({{1}, {2}, {4}, {1, 2}, {1, 4}, {2, 4}, {3, 4}, {1, 3, 4}, {2, 3, 4}} , ⊆)
Let X is the number of minimal elements in the POSET and Y is the number of maximal elements in the POSET,
Then |X-Y| is____________

Q7 Consider the following POSET: ({2,3,4,9,18} , |)
Which of the following is/are true with respect to above POSET?
(A) Maximum element does not exist in the POSET.
(B) Exactly four pairs of elements in the above POSET are not comparable.
(C) Least upper bound of 3 and 9 is 18.
(D) Number of minimal elements and number of maximal elements are same.

Q8 Let A be a set with |A| = n, and let R be an equivalence relation on set A with |R| = r. Which of the following is/are TRUE?
(A) r – n will always be even.
(B) r – n will always be odd.
(C) r – n may be zero.
(D) r – n may be 1.

Answer Key

Q1	**9~9**
Q2	**15~15**
Q3	**24~24**
Q4	**35~35**
Q5	**(B)**
Q6	**0~0**
Q7	**(A, D)**
Q8	**(A, C)**

Hints & Solutions

Q1 Text Solution:

Cardinality of equivalence relation on set A induced by partition {{a, c}, {d}, {b, e}} of set A = |{a, c} x {a, c}| + |{d} x {d}| + |{b, e} x {b, e}| = (2x2)+(1x1)+(2x2) = 9

Q2 Text Solution:

The cardinality of largest partial order relation possible on set A is 15.

Q3 Text Solution:

The number of total order relations possible on set A = 4! = 24.

Q4 Text Solution:

The number of partitions of set A such that equivalence relations induced by those partitions will contain exactly 12 ordered pairs are 35.

Q5 Text Solution:

Let

A = {1,2,3}

R = {(1,1), (2,2), (3,3), (1,2), (2,1)} ---- Equivalence

S = {(1,1), (2,2), (3,3), (2,3), (3,2)} ---- Equivalence

R ∪ S = {(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2)} ---- Not Equivalence as the transitive property is failing due to absence of (1,3) and (3,1)

Q6 Text Solution:

The minimal elements are {1}, {2}, {4}. Hence, X = 3.

The maximal elements are {1,2}, {1, 3, 4}, {2, 3, 4}. Hence, Y = 3.

The value of |X-Y| is 0.

Q7 Text Solution:

There are two maximal elements 4 and 18. Hence, maximum element does not exist in the POSET.

The five pairs that are non-comparable are (2,3), (2,9), (3,4), (4,9), (4,18).

Least upper bound of 3 and 9 is 9.

Number of minimal elements and number of maximal elements are same which is 2.

Q8 Text Solution:

r – n will always be even.

r – n may be zero.